

GENERAL SPECIFICATION G9.1

Environmental Considerations and Protection

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G9.1 ENVIRONMENTAL CONSIDERATIONS AND PROTECTION

G9.1.1 General

- (a) The Contractor shall be responsible to identify, manage and mitigate all impacts on the environment which results from his construction activities. Such impacts include any form of pollution and excessive noise affecting those outside the site boundary.
- (b) The Contractor shall also be responsible for ensuring the health of the public who may be affected by his construction activities.
- (c) The Contractor shall comply with all relevant Acts, Regulations and Codes of Practice of Singapore including any amendments or re-enactment thereto including but not limited to:
 - (i) Code of Practice on Environmental Health;
 - (ii) Code of Practice on Pollution Control;
 - (iii) Code of Practice on Surface Water Drainage;
 - (iv) Code of Practice on Environmental Control Officer;
 - (v) Guide Book on Silt Control on Construction Sites;
 - (vi) Handbook of Scope of Works for Mosquito Control;
 - (vii) Environmental Guide Book for Construction Sites;
 - (viii) Code of Practice for Noise Control on Construction and Demolition Sites; and
 - (ix) Code of Practice on Environmental Control Officer.
- (d) The Contractor's environmental team shall comprise of sufficient workers for the duration of the Project, solely for the purpose of environmental control and maintenance only. This team of workers shall not be employed to work as part of the construction team. These environmental works shall be easily identifiable by attire; e.g. a different safety vest or helmet.
- (e) The Contractor shall include the requirements of this specification in its rates and prices and no claim by the Contractor for complying with the provisions of this Specification shall be entertained by the Board.

G9.1.2 Submission of Data

- (a) The Contractor shall submit operating and pollution data for his proposed plant and equipment when required by the S.O..
- (b) The Contractor shall submit an Environmental Management and Monitoring Plan (EMMP) that includes site-specific environmental management plans which shall include as a minimum:
 - (i) Air Pollution Control Plan;
 - (ii) Vector Control Plan;
 - (iii) Waste Management Plan;
 - (iv) Noise Management Plan;

- (v) pWater Pollution Management Plan; and
- (vi) Environmental Risk Register with environmental risks and mitigation measures.
- (c) The EMMP shall take into consideration the relevant regulations and Codes of Practice. The EMMP shall be submitted to the S.O. for approval within 60 days of Commencement.
- (d) The Contractor shall continuously review and revise the EMMP as required.
- (e) The Contractor shall put in place additional measures and resources as required by the S.O. if current measures or resources are deemed insufficient. The Contractor shall deem to have considered means and included all costs to ensure that their operations are conducted in compliance to local environmental regulations and in an environmentally responsible manner. Any variation claims or claims for extension of time or will not be permitted.
- (f) The Contractor shall submit operating and pollution data for his proposed plant and equipment when required by the S.O..
- (g) The Contractor shall also maintain and make available resource usage data of the project.
- (h) The Contractor shall be responsible for the accuracy of the data and auditable records shall be kept for verification or as requested by the S.O..
- (i) The Contractor shall submit the data as part of the monthly HSE Report.

G9.1.3 Environmental Control Officer (ECO)

- (a) Environmental Public Health Requirements

The Contractor shall comply with all the provisions of the Environmental Public Health (Employment of Environmental Control Officers) Order 1999, the Environmental Public Health (Qualifications of Environmental Control Officers) Notification 1999 and the Environmental Public Health (Registration of Environmental Control Officers) Regulations 1999.
- (b) Environmental Control Officer (ECO)
 - (i) The Contractor shall engage a full-time ECO registered with the Commissioner of Public Health.
 - (ii) The ECO shall discharge his duties set out in the Code of Practice for ECO and advise the Contractor in the following main areas:
 - 1) Control of disease-bearing vectors and rodents;
 - 2) Proper management and disposal of solid waste and liquid waste;
 - 3) Control of noise and dust pollution;
 - 4) Drainage Control;
 - 5) General Housekeeping; and
 - 6) Earth control measures and silt control.

- (iii) Full-time ECOs are required to spend at least 40 hours a week. Full-time ECO shall not be allowed to take the role of the foreman or vice versa.
- (iv) The ECO shall pay attention to earth control measures and silt control of the worksite including site checks and submission of reports on alternate days or more frequently if required.
- (v) ECOs shall use the checklist provided by the Qualified Erosion Control Professional (QECF) to carry out the ECM checks. ECOs shall also highlight problems relating to ECM to QECF so that reviews to the ECM can be done on a timely basis.
- (vi) The Contractor shall be deemed to have allowed for complying with all relevant current requirements in his prices, regardless of whether those are different from the requirements specified in in this specification.

G9.1.4 Climatic and Tidal Conditions

- (a) In planning the control measures necessary to minimize environmental pollution, the Contractor shall take into account the climatic conditions in Singapore. Detailed statistics can be obtained from the Meteorological Service Division, National Environment Agency (NEA).
- (b) Where the state of the tide would affect the control measures being implemented to minimize environmental pollution, the Contractor shall make reference to the tidal information available from the Maritime and Port Authority (MPA) of Singapore.

G9.1.5 Air Pollution

- (a) The Contractor shall ensure that any atmospheric pollution (in the form of smoke, fumes, vapours, dust and other pollutants) shall be effectively controlled at the site. Burning of materials on the site will not be allowed. Flammable debris and refuse (wood or others) shall be removed from the site.
- (b) Air pollution control requirements such as the concentration and rates of emission of air pollutants shall be within legal limits. The S.O. reserves the right to request for a newer machine or an emission control device be installed if any machine or plant is deemed to be producing excessive greyish/black smoke.
- (c) Air quality on site shall be monitored on a frequency deemed appropriate by the S.O. or the ECO. Gases to be monitored include, but not limited to, carbon dioxide, methane (CH₄) and other flammable gases, with emergency plans prepared in the event permissible limits are breached.
- (d) All dusty operations must be adequately shielded and/or arrested with water at point of impact.
- (e) Pave all vehicular access with suitable materials such as concrete, mill waste or hardcore.

G9.1.6 Water and Land Pollution

- (a) No trade effluent other than that of a nature or type approved by NEA Director-General shall be discharged into any watercourse or land.
- (b) The Contractor shall ensure that all activities involving repair, servicing, engine overhaul works etc. shall be carried out on a concreted area which shall be bunded or scupper drains provided to channel all wastewater into the sewerage system. Oil removers/interceptors shall be provided to treat oil waste from workshop areas.
- (c) Diesel drums and chemicals shall be stored under shelter within concrete bund walls or in storage containers with good ventilation. Spill trays shall be provided for all drums, plants and machinery and potentially pollutive substances used on site. Spill trays shall be regularly maintained to prevent rain from washing out the pollutive substances.
- (d) The Contractor shall put in place a response plan to cater for accidental spillages into any watercourse. This plan shall be communicated to all project personnel.
- (e) The Contractor shall conduct an emergency spillage exercise at least once per year.
- (f) Emergency spill kits shall be provided on site in the event of any chemical spillages. Emergency response team shall also be competent in the use of these spill kits.
- (g) All accidental spillages and trade effluent discharges shall be investigated and reported to the Board.

G9.1.7 Rodent Control

- (a) The Contractor shall ensure that rodents are controlled at the site. Source reduction shall be used as the main form of rodent control. Food consumption and storage must be strictly restricted to designated areas where lidded rubbish bins are available.
- (b) Food must only be stored in rodent proof storage containers/ cabinets with at least 60cm clearance above the ground.
- (c) In-house pest control team should also look out for evidences of rodents and their burrows during their rounds.

G9.1.8 Vector Control

- (a) The Contractor is required to implement comprehensive vector surveillance and control at the site, including all necessary measures to prevent the site from becoming favourable to the breeding and harbouring of vectors. Accordingly, the Contractor shall undertake but not limited to the following:
 - (i) Within 3 months of Commencement, the Contractor shall submit a detailed plan for vector control. The proposal shall include the details of the Pest Control Operator (PCO) with their track record and the proposed surveillance and control measures for this site. Chemicals to be used for treatment are subject to S.O.'s approval.

- (ii) The Contractor shall engage an external NEA-licensed PCO to supplement the in-house vector surveillance and control. The PCO shall carry out vector control and surveillance at least once a week. Additional PCO services will be required during epidemic periods and incremental costs to provide for additional PCO services are deemed to have been included.
- (iii) During the construction period, the Contractor or his hired vector control company must maintain a site register which gives an up to date account of surveillance and control work that has been carried out. This register must be made readily available upon request.
- (iv) Source reduction and effective drainage shall be used as the main forms of mosquito control. Environmentally friendly applications such as the use of *Bacillus thuringiensis israelensis* (Bti) should be used as far as possible to supplement source reduction.
- (v) Pest control operators must be NSRS (National Skills Recognition System) certified in pest control.
- (vi) Contractor must have an in-house pest control team to search and destroy any potential breeding grounds, especially after every rainfall, using the "zoning method". Personnel involved in vector surveillance and control shall undergo relevant trainings which include:
 - 1) Joint ITE-NEA Certificate in Pest Management; and
 - 2) Trainings on understanding vector-borne diseases identifying potential vector breeding grounds and measures to prevent the propagation of vectors for general workers.
- (vii) Monthly returns of the site records are to be submitted to the respective Environmental Health Offices.
- (viii) Provide other information or comply with amendments to the forms as may be requested from time to time by the S.O..
- (ix) The Contractor shall ensure source reduction such as larvicide or Bti are administered to all potential areas of breeding (especially earth retaining structures such as walers and struts).
- (x) The in-house pest control team shall carry out search and destroy activities of any potential breeding grounds, especially after every rainfall, using the "zoning method". The team shall:
 - 1) Divide the construction site into appropriate number of zones for vector control particularly for mosquito control;
 - 2) Carry out vector surveillance and control activities in at least one zone per day; and
 - 3) Ensure that sub-contractors are carrying out proper housekeeping at their individual work zones to complement the in-house pest control team effort.

- (xi) The Contractor shall regularly monitor the adult mosquito population using well-maintained gravitraps. Gravitraps shall be placed in an area with good housekeeping, under shade and on a levelled ground for it to be effective. The records shall be documented and submitted to the relevant agency when requested.
- (xii) Thermal fogging shall only be carried out when there is high population of adult mosquitoes and/or when there are dengue/zika outbreaks near the construction site in accordance to the Code of Practice on ECO. The Contractor shall ensure that the PCO has obtained approval from NEA before fogging is carried out on site.
- (xiii) All site offices/containers must have a sloping/pitched roof installed with the sides adequately shielded from rain. Containers for office or storage purposes on site shall be sited on concrete paved ground with perimeter drains for effective surface water drainage.
- (xiv) The Contractor shall ensure that no puddles of water are formed on the ground by using appropriate cover such as concrete paved, milled waste or steel plates.
- (xv) The Contractor shall prepare a vector outbreak response plan as a part of the overall vector control plan and shall immediately report all suspected vector outbreaks such as dengue and zika to the Board and S.O..
- (xvi) In the event where mosquito breeding is discovered on site by NEA, the Contractor is to carry out a Vector Control Time-out. The Time-out is to involve a detailed search and destroy effort to eliminate any potential breeding grounds
- (xvii) The Contractor shall implement a system of tracking and maintaining an updated list of all personnel entering the site including name, nationality and contact information.
- (xviii) Any person found on site to be a carrier of the dengue haemorrhagic fever virus, malaria or Japanese Encephalitis or any other vector-borne disease shall be immediately removed from site to appropriate medical care to prevent the outbreak of the disease. The Contractor shall develop a response plan specifying the measures to be taken in the event there is any person found on site to be a carrier of or contracted with any vector-borne disease.

G9.1.9 Waste Management

- (a) The Contractor is required to identify all types of waste (e.g. construction waste, chemical waste, wood waste, metal waste, plastic waste, office waste, etc) and implement comprehensive waste management system at the site in order to ensure minimum wastage, proper disposal and prevent pollution to the environment.
- (b) The Contractor shall provide adequate number of bins of capacity not less than one cubic metre at the site for the storage of all inorganic waste such as building debris, dust, dirt and litter. Separate scrap metal waste bins may also be provided.
- (c) An adequate number of bins with air-tight covers of not less than 85 litres shall also be provided for the storage of organic waste.

- (d) The Contractor shall clear all site waste as often as may be necessary to prevent build-up and deposited/stored in these bins. The bins shall be removed from site and replaced/emptied once they have been filled.
- (e) The Contractor shall carry out effective on-site sorting of construction and demolition materials to recover inert, reusable and/or recyclable portion.
- (f) The system of on-site sorting and temporary storage of construction and demolition materials shall include the following:
 - (i) Metals shall be recovered for collection by recycling contractors;
 - (ii) Cardboards and paper packaging shall be recovered, properly stockpiled in dry and covered conditions to avoid cross contamination by other construction and demolition materials; and
 - (iii) Excavated materials shall be sorted to recover inert portions (e.g. soil and crushed rocks) for re-use on site or disposal to designated filling areas.
- (g) All construction debris shall be disposed of at the gazetted Government dumping grounds or at such other sites or locations as shall be directed by NEA. Disposal of domestic refuse may be arranged with the Environmental Public Health Division (EHD). The Contractor shall pay all tipping fees at the gazetted dumping grounds.
- (h) Any toxic waste as listed in the Environmental Public Health (Toxic Industrial Waste) Regulations and this includes used oil, shall be disposed of by an operator holding a valid toxic industrial waste collector's license.

G9.1.10 Earth Control Measures (ECM)

- (a) The Contractor shall be responsible for preventing silt from being washed into public drains and the canal by implementing effective Earth Control Measures (ECM) for the construction site to meet the requirements under the latest Sewerage and Drainage Act Cap 294. (Information on ECM requirements can be found in the Code of Practice on Surface Water Drainage and the website www.pub.gov.sg/ECMdrainage/earthcontrolmeasures).
- (b) The Contractor shall implement effective Earth Control Measures (ECM) during construction at all times. The ECM shall be implemented according to the Code of Practice for Surface Water Drainage. The Contractor shall note that the ECM are meant for the containment and treatment of silty rainwater runoff only and not meant for the treatment of process water from construction activities such as slurry from tunnelling, pipe-jacking and bore-piling works. Such process water should be handled and treated to comply with the requirements under the Environmental Pollution Control Act (Chapter 94A).
- (c) Within 2 weeks of Commencement, the Contractor shall submit the schematic ECM plans of the construction site for the whole duration of the Contract period taking into account the various ECM requirements under the different phases of the construction activities. The Contractor shall engage a QECP to design and endorse the ECM plan. The Contractor shall implement ECM according to the QECP's ECM plan before earth works start. During the course of construction, the Contractor shall engage his QECP to review the effectiveness of the ECM regularly, in tandem with the various phases of construction works. The Contractor shall revise the ECM as and when advised by his QECP.

- (d) The Contractor shall maintain the ECM for the whole duration of the Contract and make necessary revisions as a result of changes in site conditions for the whole duration of the Contract to ensure that it is effective at all times. Proper records detailing the maintenance works, supported by dated photographs, shall be kept by the Contractor for verification.
- (e) The Contractor shall keep good records of the ECM operation and maintenance.
- (f) The Contractor shall not remove the ECM until all works are completed and upon the advice of his QECP. Records of approval for removal of ECM by QECP shall be provided upon request by S.O..
- (g) The Contractor shall submit the ECM proposal duly endorsed by his QECP, in the format prescribed by the Board, to Director, Catchment & Waterways (PUB) to obtain the necessary clearance before earth works start. The proposed ECM plan shall include the following content:
 - (i) Project Brief
 - 1) Project description;
 - 2) Name and address of site occupier;
 - 3) Site area and Contract period;
 - 4) Location map and site plan; and
 - 5) Construction phases and schedules.
 - (ii) ECM Design Calculations.
 - (iii) Erosion Control Plan.
- (h) The Plan at the very least shall depict graphically the activities, including sequence of work, type and duration for each phase of construction activities to include the following measures to:
 - 1) Sequence and schedule of the earthworks/demolition works in stages and progressively with the subsequent construction activities and building works.
 - 2) Minimise site disturbance by keeping site clearance works to a minimum by retaining as much of the existing vegetation as possible.
 - 3) Pave up the bare surfaces and all construction access using concrete or milled waste or other suitable materials.
 - 4) Protect the bare slopes with close-turfing, concrete grouting, canvas or erosion control blanket.
 - 5) Protect the earth stockpiles with canvas or erosion control blanket. Earth stockpiles shall be carted away if not drawn down within 3 months.
 - 6) Restore ground cover over disturbed areas which have become bare immediately to prevent soil erosion.
 - 7) Trench excavation works shall be carried out in sequence with the progress of permanent works to minimise impact on the environment.
 - 8) Cover up the active work surfaces with canvas sheet during rain or at the end of the workday.

- 9) The Contractor shall identify the exposed bare surfaces and slopes for turfing or paving operation before the start of each phase of the project.
- (ii) Sediment Control Plan
- 1) Concrete-lined cut-off drains (minimum C7 precast channel) along perimeter of the construction sites.
 - 2) Sealing of site hoarding.
 - 3) Ramp/curb at site or work zone entrance/exit.
 - 4) Separate drainage for bare earth surfaces and non-bare earth surfaces.
 - 5) Silt fence properly installed and embedded onto the ground along the internal and perimeter cutoff drains. The silt fence shall be installed using an approved geotextile filter fabric to capture sediment from the storm water run-off. The sediment built up behind the silt fence must be regularly removed.
 - 6) Intermediate silt traps of suitable size shall be installed at regular intervals along the perimeter lined cut-off drain. Within the intermediate silt traps, suitable geotextile filter fabric or equivalents shall be installed across the full depth and width and / or coagulation-assistance materials shall be placed. Silt traps relying primarily on hardcore, granite chips or sands for filtration, are not acceptable.
 - 7) Holding sumps/ponds for silty water. These shall be treated and discharged within 10hr after each storm event.
 - 8) Adequate silty water treatment systems to treat silty water before the discharge into the public drain or waterbody. The silty treatment systems shall be sized to ensure that the silty water in the holding sumps/ponds can be treated and discharged within 10hr after each storm event.
 - 9) Turbidity curtains for works in or adjacent to water bodies, such as canals, rivers, sea or in reclamation works. Turbidity curtains shall be installed along all the exposed slopes/riverbanks for works in or adjacent to water bodies, such as canals, rivers, sea or in a reclamation work. The silty discharge trapped within the turbidity curtain shall be allowed to be settled or treated and removed.
- (iii) Site ECM Management System
- 1) Daily ECM inspection report by the QECP appointed site personnel.
 - 2) Regular site inspection by the QECP to review the ECM plan.
 - 3) TSS monitoring device for continuous monitoring of the TSS of the discharge into the storm water drainage system. The Contractor shall submit a report on the weekly TSS concentration to the Board and the S.O. once every 2 weeks.

- 4) The Contractor shall install a CCTV monitoring system and with continuous real time “live” video feed of the discharge and TSS for every final discharge point. The information shall be made available on the internet. This system should consist of an “alert” feature when allowable limits are exceeded and notifications shall be immediately provided to the S.O. and others to be designated by the S.O.. The information from the real time monitoring shall be made accessible and available to the S.O. at all times including through the S.O. CCTV. The Contractor shall take reference to Annex 9 of the Board’s ECM Guidebook ([www.pub.gov.sg /Documents/ECM_Guidebook.pdf](http://www.pub.gov.sg/Documents/ECM_Guidebook.pdf)) as the minimum requirements. The Contractor shall submit the monitoring system proposal to the S.O. for acceptance.
- (i) The QECP shall submit the full basis of design complete with design calculations to show clearly how the perimeter cut-off drains, silt fence, sedimentation basins and storage basins/tanks are designed and sized as well as how such design will effectively filter off silt and allow only clean water to discharge into public drains or water bodies. The QECP shall also submit the design specifications on the treatment system which can effectively treat the silty discharge from the construction site. The intensity of a one-in-five year storm should be adopted for the proposed design. More conservative storm options can be considered by QECP to ensure that the ECM measures are adequate and these assumptions shall be presented in the design submission. The submission of the basis of design will not absolve the QECP and Contractor of the responsibility of ensuring that silt will not be washed into public drains or water bodies.
- (j) Notwithstanding the above general specifications, the following requirements must be taken into consideration by the QECP in the preparation of the Erosion and Sediment Control Plans:
- (i) Pervious and impervious areas of the Contractor’s site office area shall be coordinated with the Contractor’s drainage plans.
- (ii) All temporary access for vehicles and construction equipment, i.e. cranes, excavators, etc. must be properly constructed. The access may be formed with well-compacted hardcore blinded with sand and concrete paved or covered with milled waste. Wheel washing facility shall be provided at exits from the worksite to fully clean all vehicle wheels of dirt, mud and construction debris and the wash water treated. The Contractor shall maintain all temporary accesses and carry out periodic topping up of well compacted hardcore or mill waste where required by the S.O..
- (iii) All earth slopes and surfaces adjacent to the worksite, which are susceptible to erosion, shall be concrete lined or covered with erosion control blanket or close turfing. The Contractor shall ensure all excavated materials and spoils are either removed from site or stored in proper containment by the end of the day.
- (iv) As far as possible, vegetation within the site shall be retained.

- (v) During trench excavation, the maximum length of trench opened should not exceed 50m. Any excavated materials stockpiled on site must be covered with canvas sheet or erosion control blanket. If the stockpile has to be left on site, a silt fence must also be constructed around the stock pile to retain any silty water that may wash out during a storm event. At the end of each work day, both backfilled and open trenches must be covered by canvas sheets firmly pegged to the ground. The backfilled trenches must remain covered until the completion of close turfing/reinstatement work. If the earth stockpile is to be left on site for more than 3 weeks, it shall be covered up with erosion control blankets or close turfing.
- (vi) A low concrete bund wall shall be constructed around the shafts to prevent ingress of storm water. At the end of each working day, the shaft area shall be covered with canvas and firmly pegged to the shaft/ground. When the excavation reaches the base level, the base shall immediately be concrete paved.
- (vii) At every 50m length of cut-off drain (or part there-of), one standard silt trap shall be provided. The standard silt trap or approved equivalent shall be of size 1m width x 2m length x 1.5m depth with suitable graded stones and geotextile or equivalent (as shown in Drawing No. 8 of 'Code of Practice on Surface Water Drainage') and polymer blocks shall be used to aid in the silt settlement before the water is discharged.
- (viii) After any rain event, the silty water pumped from the trenches, shafts, sedimentation ponds, within the silt-fenced areas, etc. must not be discharged into the open drain. Instead it should be stored in portable tanks for treatment by silty water treatment unit or transported to pre-approved, designated water treatment plant for treatment to meet the discharge standard.
- (ix) QECP shall provide a site specific ECM checklist for the Environmental Control Officer (ECO) to conduct daily checks on ECM for the worksite and the ECO shall submit the checklist to all relevant parties, especially the QECP, upon completion of each inspection. The QECP shall follow up on the ECO's inspection and ensure that the relevant ECM issues are addressed by the Contractor. Where necessary, the ECO's comments will also be used as basis to review the ECM for the project. A log book shall be kept on site for QECP and ECO to put down their observations or instructions after every inspection.
- (x) QECP shall conduct weekly site checks and submit endorsed report with photographs, assessments and recommendations throughout the whole duration of the Works. On making these site visits, the QECP shall maintain a log of his visits and report to the S.O. or his representatives during these visits the findings and proposed resolutions of the infringements (if any). The QECP shall coordinate with the S.O. or his representatives for joint site inspections where required.

- (xi) The above requirements are deemed to be included in the items for the implementation and maintenance of Erosion and Sediment Control Plans. The Contractor is advised to price adequately for these items. Notwithstanding the provision of the above-mentioned measures, Contractor shall note that adequate ECM must be provided throughout the project duration. Submission of ECM plans to the S.O. does not absolve the Contractor of the need to ensure that the Works comply with the Code of Practice on Surface Water Drainage.
- (k) Contractor shall have at least one of the full-time site staff equipped with Earth Control Measures Officer (ECMO) qualification and be registered as a ECMO under the IES (The Institution of Engineers Singapore) ECMO Registry to perform the following roles:
 - (l) The ECMO's role shall include:
 - (i) Implement ECM plan according to QECP design;
 - (ii) Devise ECM operation;
 - (iii) Maintenance and inspection of ECM programme on site; and
 - (iv) Assist the QECP to monitor ECM effectiveness throughout various stages of construction.

G9.1.11 Turbidity Curtain

- (a) For water bodies which are raw water sources for potable water and/or need for recreational purposes, high turbidity of the water in the water body will affect the treatment costs for potable water and/or the recreational use.
- (b) The Contractor shall install turbidity curtains for works in such water bodies and any other water bodies specified elsewhere. The works including the design, fabrication, and installation of the primary and secondary turbidity curtains shall ensure that the turbidity of water in the water bodies around the works shall not exceed the pre-existing levels or 50 mg/L, whichever is greater.
- (c) The works in this Specification consists of all construction operations relating to the turbidity curtain. These construction operations include, but are not necessarily limited to the following:
 - (i) Design, manufacture, install and maintain primary and, if necessary, secondary turbidity curtain(s) around the areas of marine construction, either across the water body to enclose the entire work area, or individual curtains within/around/along the water body to enclose work areas.
 - (ii) Curtain(s) shall remain in place during excavation, installation of piles, foundations etc. and backfilling works.
 - (iii) Maintain the curtain(s) in good working condition for the duration of construction works.
 - (iv) The turbidity curtain(s) shall not be removed until all operations have been completed and the water quality within the confines of the turbidity curtain meets the standards. Install as many turbidity curtains as are necessary to meet the water quality standards.

- (v) The primary turbidity curtain shall be installed as specified. The secondary curtain (s) shall be installed as and when necessary to meet the requirements of these specifications.
- (d) The purpose of the curtain(s) is to meet the water quality standards by minimising the transport of silt and other constituents generated by construction activities in the water body. This includes excavation, wet recovery of micro tunnel equipment, bentonite slurry use, tremie concrete operations, backfill and all other construction activities conducted in or near the water body. The turbidity curtain system shall provide sufficient residence time to allow soil or bentonite slurry particles to fall out of suspension, reduce turbidity, and meet the water quality standards.
- (e) Since it will require time to install additional, secondary turbidity curtain(s), the Contractor shall take turbidity measurements at specified distances from the edge of each outermost installed curtain to allow time to install secondary curtain(s) before the turbidity limit is exceeded.
- (f) Develop a plan to monitor the turbidity throughout the water column at three distances from the edge of each outermost installed turbidity curtain. Make initial measurements at a distance of 30m, 60m, and 100m from the edge of the primary curtain with turbidity measured at four depths in the water column, then adjust distance and depth as appropriate based on direction of the plume, plume velocity, and the change in turbidity with distance from the work area. Initial measurement shall be made three times a week.
- (g) Have the approved design of secondary curtain(s) available before starting the excavation. Furthermore, for each area surrounded by a primary turbidity curtain, have at least one secondary curtain available onsite and ready for installation after the turbidity limit is exceeded at a distance of 60m from the edge of the installed curtain.
- (h) The above requirements shall be met before construction activities begin.
- (i) Prior to manufacturing the primary and secondary curtain(s), the Contractor shall submit the details of the primary and secondary curtain fabrication including:
 - (i) Material certifications and data on physical properties and ultraviolet resistance of permeable and impermeable curtain fabrics;
 - (ii) Shop Drawings for curtain and appurtenances;
 - (iii) Design analyses and calculations;
 - (iv) Installation plan and configuration;
 - (v) Flotation and anchoring plan;
 - (vi) Maintenance plan;
 - (vii) Methods for providing entry and exit through curtain(s) as necessary for construction of all offshore work;
 - (viii) Manufacturer/Supplier qualifications; and
 - (ix) Profile of water body bed along curtain alignments.

- (j) The Contractor shall use the following requirements as guidelines for the curtain fabric:
 - (i) Primary Turbidity Curtain
 - 1) Curtain Section: Curtain shall be a combination of permeable or impermeable materials. Curtain shall be heavy-weight, flexible, nylon-reinforced, polypropylene filter fabric, or flexible nylon reinforced thermoplastic as necessary to control turbidity created during construction, sewn into panels, hemmed, and edges finished to prevent ravelling.
 - (ii) Secondary turbidity curtain(s)
 - 1) Curtain Section: As specified for primary curtain or as necessary to control turbidity near the construction.
 - (iii) Connectors
 - 1) Provide the curtain with appropriate galvanized steel snap hooks and rings for connecting load lines.
 - (iv) Flotation
 - 1) Provide a sufficient number of floats to keep the top of the curtain above the water surface by a minimum of 150mm at all times.
 - (v) Ballast and Anchorage
 - 1) Each curtain shall be equipped with a galvanized steel chain integrated into the bottom of the fabric to keep the curtain vertical and in contact with bottom of the water body. Each curtain shall also be anchored to the water body bed to prevent excessive displacement from wind, waves and currents. The ballast, anchorage, and floatation shall be designed by the curtain manufacturer for the wind and wave conditions, bottom profile, and changes in water level. Anchors shall be spaced as necessary to secure each curtain and keep it stable in all conditions. Anchorage and/or flotation shall be designed to keep the top of each curtain above the water surface when subjected to wind or wave forces.
 - 2) Design, provide, and install shore anchoring where each curtain is attached to the shoreline. Design, provide, and install marine anchorages as necessary to secure each curtain.
 - (vi) Load Line
 - 1) Fit the curtain(s) with galvanized wire rope with vinyl coating of sufficient strength to resist all internal and external loading.
 - (vii) Oil Booms
 - 1) Oil booms, skimming devices, and pollution containment devices shall be provided as and when necessary to prevent contamination of the water.

(viii) Pre-installation Profile

- 1) Prior to manufacturing the curtain(s), develop a profile of the water body bed for each curtain location. Verify the depth of curtain, especially if the curtain is to be anchored to the shoreline to confirm the bottom profile at the exact location of curtain placement.
- 2) The curtain manufacturer shall use this information to dimension the curtain(s) with allowances for water level changes.

(ix) Curtain Design

- 1) The curtain and oil boom systems shall consist of a primary curtain, plus secondary curtain(s) as necessary to meet the water quality standards specified.
- 2) Each primary curtain shall have a section that is full depth and is in contact with the bottom of the water body.
- 3) Each secondary curtain, if used, shall consist of a section that extends up to a height as necessary to control turbidity.
- 4) Design each curtain for a useful life of at least the duration of the project.
- 5) Design curtain system for all temperature, wind, wave and current conditions at the project site as well as the anticipated varying water levels.
- 6) Design curtain system to meet the water quality standards.

(x) Curtain Fabrication

- 1) Each curtain shall be manufactured / supplied by a specialty subcontractor with experience in turbidity curtain design and fabrication.
- 2) Design curtain to accommodate expected water level variations. If necessary, provide each curtain with additional longitudinal panels that can be added when the water level rises and with removable panels that can be pulled out when the water level drops.
- 3) Access Gate: Provide means for movement of equipment or materials through the curtain as may be required for operations.

(xi) Primary Curtain Installation

- 1) The exact locations shall provide sufficient working space compatible with the construction methods and within the work limits of the Contract. Install primary curtain before commencement of any works in or around the water.
- 2) Deploy the curtains in conformance with the manufacturer's recommendations.

(xii) Secondary Curtain Installation.

- 1) If the turbidity measured 60m from the primary curtain exceeds the water quality standards, install secondary curtain(s).
- 2) Have the assembled curtain materials on site and install secondary curtain within 2 days of the day the turbidity exceeds the water quality standard.

- (xiii) Maintenance
 - 1) Maintain, repair, and adjust the curtains as necessary throughout all construction activities.
 - 2) Visually inspect the turbidity curtain(s) at least weekly. A written copy of the inspection report shall be submitted.
- (xiv) Curtain Removal
 - 1) At the completion of all construction activities remove all turbidity curtains in their entirety. This includes all anchoring devices.
 - 2) Do not remove the curtains until the water inside the enclosed area meets the water quality standard.
 - 3) Obtain approval before removing curtains.

G9.1.12 Flooding

- (a) The Contractor shall take into consideration of possible flooding to the Works during the construction stage. The Contractor shall consider the risks of flooding due to nature or arising from the construction activities.
- (b) The Contractor shall prepare a comprehensive 'flood protection plan' for the whole of his Works and submit to the S.O. for his acceptance within 2 months of the Commencement of the Contract.
- (c) The plan shall identify the areas that are at risk from flooding. The plan shall examine the potential consequences of any flooding and shall make proposals to prevent flooding.
- (d) The flood protection measure provided shall comply with the requirements of "Code of Practice on Surface Water Drainage" issued by PUB Catchment & Waterways Department.
- (e) When pumping is proposed, the Contractor shall address the following:
 - (i) Operating procedures;
 - (ii) Provision of a secure power supply and back-up;
 - (iii) Automation and/or reaction time required to start the pumps; and
 - (iv) Method and arrangement of discharge.
- (f) The Contractor must make adequate safety provisions in the event of flooding whether it is due to inclement weather or the river bank bursting to minimize the loss of life and property damage, where practicable.
- (g) The safety provisions to be put in place shall not be limited to providing of an emergency evacuation and rescue plan, trained rescue teams and water rescue aid or equipment. The trained rescue team and emergency procedure shall be in place before the start of any work. The number of trained rescue teams and water rescue aid or equipment shall be increased when deemed necessary by the S.O. or his representative. The water rescue aids or equipment shall be properly maintained throughout the entire duration of the project.

G9.1.13 Contaminated Grounds

- (a) The Contractor shall not contaminate any ground arising from his Works. The Contractor shall take all necessary precautions to prevent any person from being exposed to the risk posed by any contaminants at the site.
- (b) Soil samples at appropriate depths and spacing from the Works shall be taken for testing by a competent person to identify any contaminant present as well as monitoring of airborne contaminants through air sampling. Soil contaminants to consider shall include, but not limited to, heavy metals and their derivatives, hydrocarbon compounds and asbestos. Airborne contaminants to consider shall include, but not limited, methane (CH₄) and other flammable gases.
- (c) It shall be noted that the contamination can be caused by a range of different hazardous materials and by a mixture of concentrations and distributions. Contaminants can also migrate to adjacent areas via ground water. The Contractor shall ensure that:
 - (i) Measures are taken in the design to alleviate the risks and describe those hazards that are not reasonably practicable to remove by design but must be controlled by the Contractor and any other relevant party;
 - (ii) Sufficient information is provided on the nature, extent and level of contamination and that all personnel entering the site shall be informed of the hazard and the precautions needed; and
 - (iii) All permissible exposure limits shall be established and communicated.
- (d) The Contractor shall comply with all the statutory requirements as laid down by NEA regarding the disposal of contaminated waste material.
- (e) The Contractor shall transport waste material directly from the excavation to the approved dumping ground and shall discharge the waste directly into position. No stockpiles of waste are allowed on site.

G9.1.14 Noise

- (a) General
 - (i) The Contractor shall ensure that excessive noise is avoided at all times to protect nearby residents as well as site personnel. The Contractor shall comply with the Environmental Protection and Management Control of Noise at Construction Sites regulations with relevant Schedules and follow the guidelines of the Singapore Standard 602:2014 – Code of Practice for Noise Control on Construction and Demolition Sites”.
 - (ii) Using recommendations in SS 602:2014, the Contractor shall submit a Noise Management Plan (NMP) as part of the EMMP outlining but not limited to the following:
 - 1) Comparison charts between baseline monitoring results and anticipated noise emission levels;
 - 2) Machinery and plant to be utilised on site as well as their noise emission levels;

- 3) Sequence of work and construction methods involved, indicating anticipated noise levels accompanying each type of activity;
 - 4) Proposed noise mitigation measures along with the estimated noise reduction levels;
 - 5) Site utilisation plan;
 - 6) Scheduling of works demonstrating consideration of noisy activities;
 - 7) Contingency measures to avoid excessive noise for works to be carried out after 7pm; and
 - 8) Public relations strategies demonstrating close community relationship for any potentially affected members of the public (signboards, newsletters, circulars, complaint handling and investigation, rapport building with residents).
- (iii) The Contractor shall note that the construction equipment and methods of work which cause excessive noise will not be allowed to be used on site. The S.O. has the discretion to require the Contractor to take necessary precautions, whether specified herein or not, to maintain or to repair such construction equipment or to instruct their removal from site when it is determined that the noise level generated from the construction works fails to comply with regulations and standards as stated in this specification.
- (iv) Sensitive buildings must be identified and mitigation measures implemented before work commences. The Contractor shall note that the S.O. has the right to give instructions to stop any noisy work activities temporarily during school examination periods, near to any educational institutes or schools to the extent such requirement applies to the Contractor's works.
- (v) It is the Contractor's responsibility to ensure that the machine/ equipment is maintained and operating to the standards indicated in their respective specifications.
- (b) Noise Measurement
- (i) The Contractor shall provide all necessary competent and qualified personnel and suitable equipment for all measurements and recordings of noise levels.
 - (ii) Unless otherwise specified, prior to the commencement of any construction activities and major diversion works, the Contractor shall carry out a survey of noise levels for one week, on a 24-hour basis, to establish the levels of the background noise for record purposes. Locations of such noise measurements shall be as directed by the S.O..
 - (iii) Unless otherwise specified, at any time during the Contract period as directed by the S.O. and after the project is completed and opened to traffic, the noise survey or part of it shall be repeated to establish any change in the noise levels.
 - (iv) Noise measurements will be required for occupied buildings within 50m from the Contract limits. Noise levels at buildings shall be measured 1m away from the nearest façade of the building to the road and readings shall be taken from at least 3 different levels (1st storey, intermediate storey and top storey) or as directed by the S.O..

- (v) Noise levels may also be required at locations other than buildings. The Contractor shall submit a detailed report including analysis within 2 days after the measurements are completed to the S.O.. If applicable, the Contractor shall also make a traffic volume count on both carriage ways of the road(s) fronting the buildings during the period of noise measurement.
 - (vi) Noise measurements are to be taken at least 1.5m above grade without any obstructions/obstacles in the direction of measurement.
 - (vii) Unless otherwise specified, the Contractor shall install real time “live” monitoring device to monitor the noise levels for the entire construction period. In addition, the Contractor shall monitor the noise levels for the whole duration of noisy activities and night works using portable noise meter.
- (c) Noise Control
- (i) The Contractor shall while preparing the programme for the Works, consider the non-working restriction and the site layout to minimise noise as far as possible including but not limited to considering using materials and other intermediate stages of construction as noise barriers, etc.
 - (ii) While planning for the Works, the Contractor shall review the working hours and consider the effects of construction noise on personnel working in or around the site as well as the neighbourhood within proximity of the site. The Contractor shall consider the nature of the land use in the area, duration of works and the effect of lengthening works period or other nuisances which may affect the neighbourhood.
 - (iii) The Contractor shall take all practicable steps including reference to SS 602:2014 to reduce noise arising from site activities to the minimum. These steps shall include, but not limited to:
 - 1) All machinery and plant must be sound reduced prior to entering the site.
 - 2) Noisy plant must be housed in an acoustic shed.
 - 3) Noisy activities must be barricaded with portable sound barriers/panels.
 - 4) All plant/machinery/equipment must be pasted with a weather-proof sticker clearly indicating its noise emission level (at source) under normal operating condition.
 - 5) All machinery in operation must have their covers properly shut at all times.
 - 6) Noise barriers should be erected before any work commences if the site is within proximity to residences and/or sensitive buildings or as directed by the S.O..
 - 7) The Contractor shall commit sufficient effort and time into public relation works to establish good rapport with the affected community. Such activities shall be subject to the approval of the Board and S.O..
 - 8) If applicable to the location of the site, the Contractor shall note that all construction works, which generate substantial noise, shall not be carried out during the school examination periods.

- 9) The Contractor shall note that if applicable to the location of the site, the S.O. has the right to give instructions to stop construction work activities temporarily during school examination periods, near to any educational institutes or schools.
 - 10) The Contractor shall obtain an official confirmation of the examination schedules from the School Administrator including revision of schedules.
 - 11) The Contractor shall schedule his work programme accordingly to avoid delay. Any claim for extension of time will not be permitted.
 - 12) The Contractor may be directed by the S.O. to suspend works immediately due to public feedbacks of noisy activities arising from the Works. The Contractor shall take adequate measures to protect all uncompleted works. Where the relevant works are temporary suspended under this Clause, the Contractor shall deem to have included such incidents in his tender accordingly. The Contractor shall not be entitled to any extension of time and any loss and expenses incurred arising from such temporary suspension of the works.
 - 13) All compressors, generators, welding sets etc. shall be of sound reduced models fitted with properly lined and sealed acoustic covers which shall be kept closed whenever the machines are in use and all ancillary pneumatic percussive tools shall be fitted with mufflers or silencers of the type recommended by the manufacturer.
 - 14) All machines in intermittent use shall be shut down or throttled down to a minimum in the intervening periods between works.
 - 15) All pile driving shall be carried out by a recognised noise reducing system. Rotary drills and busters actuated by hydraulic or electrical power shall, where practicable, be used for excavating hard material. Noisy construction plant such as cement batching plant, shall be sited as far away as possible from occupied buildings with noise barriers erected, specifying the proposed location for the noise barriers.
 - 16) Care shall be taken when loading or unloading vehicles, dismantling scaffolding or moving materials to reduce impact noises. Access to the working areas shall be such as to ensure a minimum disturbance to persons in occupied buildings. The Contractor shall not execute any of the works or carry out maintenance of construction plant in such a manner as to cause nuisance unless the work is necessary for the saving of life or property or for the safety of the works in which case the Contractor shall immediately advise the S.O..
- (iv) For determining the maximum or overall maximum permitted noise level, the readings shall be taken:
- 1) 50 m from the boundary of the construction site if there is any building located within 50 m of that boundary.
 - 2) One metre away from the outside of the affected building if the affected building is more than 50 m from the boundary of the construction site and there is no building located within 50 m of that boundary.

- (v) If there are other sources of noise affecting the measurement of the level of noise emitted from a construction site, the maximum permissible noise level shown in the above specified regulations (Environmental Protection and Management Control of Noise at Construction Sites) shall be adjusted and the adjusted value shall be obtained by adding the correction factor corresponding to the difference between the maximum permissible noise level and the background noise level to the higher of the two noise level. The adjusted value shall be taken as the maximum permissible noise level.

G9.1.15 Control of Vibrations at Construction Sites

- (a) Before commencing construction work in the vicinity of buildings, structures and existing services, the Contractor shall verify the exact locations of these buildings, structures and existing services. The Contractor shall submit a vibration control and monitoring plan to the S.O. for approval prior to commencing such work.
- (b) The Contractor shall design both temporary and permanent works to ensure that ground movements are kept to an absolute minimum. Proven techniques and good workmanship are required to restrict ground loss that would lead to ground movement.
- (c) The Contractor shall select and operate construction equipment in a manner such that vibration induced damage to buildings, structures or existing services is negligible.
- (d) The Contractor shall be fully liable for any repair of buildings, structures or existing services damaged by the effects of his construction works. The Contractor shall also carry out any precautionary, protective or recovery measures deemed fit by the Contractor and agreed by the S.O. during the construction stage or the defects liability period of his works.

G9.1.16 Construction Near Buildings, Structures and other Services

- (a) The Contractor shall establish and inform the S.O. of the procedures and methods for all excavation and other construction work to prevent damage to adjacent buildings, structures or existing services during the Works.
- (b) Equipment vibrations – The peak particle velocity (PPV) impacted to adjacent buildings, structures and existing services shall not exceed the latest requirements set forth in British Standard BS 7385. This requirement applies to all construction equipment pipe-jacking, piling and tunnelling machinery. The Contractor shall provide a detailed impact assessment that clearly demonstrates the vibration attenuation for each specific equipment in similar ground conditions. This may require on-site testing and monitoring with a seismometer.
- (c) Load transfer from foundations to buildings, structures and existing services – Load transfer (if any) from the foundation elements to building, structure and existing services (e.g. by skin friction) shall not exceed 10 kPa (10 kN/m²). This requirement may result in the foundation elements being "debonded" at and above the building, structure and existing services.

G9.1.17 Assessment of Construction Effects on Buildings, Structures and Existing Services

- (a) The Contractor shall carry out an impact assessment of the construction effects of his works on the buildings, structures and existing services near his works and submit the Construction Impact Assessment Report to the S.O. prior to commencing the works. The Report shall include details on the following:
 - (i) The ground conditions, geotechnical profiles and relevant borehole logs;
 - (ii) The proposed construction equipment and methods, and sequence of construction;
 - (iii) Assessments to adequately address construction equipment, methodology, vibrations, ground displacements, load transfer from foundation elements to the buildings, structures and services;
 - (iv) Vibration characteristics and attenuation for major construction equipment and methodology;
 - (v) Instrumentation and monitoring requirements;
 - (vi) The damage potential and risks to the adjacent buildings, structures and other services; and
 - (vii) Preventative and remedial measures that would be utilized to protect the adjacent buildings, structure and other services.
- (b) The Contractor shall advise the S.O. as to the risk and level of damage expected to the adjacent buildings, structure and other services based on his detailed analysis and assessment. The Contractor shall propose mitigating measures to protect the buildings, structures and other services from damage, to the satisfaction of the S.O..

G9.1.18 Instrumentation and Monitoring Programme

- (a) Instrumentation and Monitoring Programme is to be provided by the Board's nominated Instrumentation and Monitoring Contractor for TWRP Project.

G9.1.19 Biodiversity

- (a) This section is only applicable to contracts with a Biodiversity Impact Assessment (BIA) conducted for their construction work.
- (b) The Contractor shall adhere to and implement the recommended mitigation measures outlined in the BIA.
- (c) The Contractor shall submit a Biodiversity Monitoring and Management Plan as specified in the BIA and submitted to the S.O. for approval within 3 months of Commencement or as subjected to the S.O.'s approval.
- (d) During construction, the Contractor shall include but not limited to the following implementation measures on site:
 - (i) Use environmentally-friendly methods for vector control such as Bti and search & destroy operation etc. instead of anti-malaria (AM) oil, or chemical larvicides;
 - (ii) Use sound reduced machines prior to entering the site;
 - (iii) Barricade noisy activities with portable sound barriers or panels;

- (iv) Lightings would affect nocturnal animals. When lighting has to be used, refrain from pointing the glare towards habitats. Lightings shall be directed downwards where work is carried out;
- (v) Site utilization plans shall consider preservation and protection of native trees as far as possible. Tree or shrubs that can be preserved and protected shall be identified with methods to prevent harm to the tree, branches and roots. (Refer to National Parks Conservation of Trees and Plants Guideline for reference); and
- (vi) Preventive measures to ensure no trade effluent, chemical, diesel or silt discharges into nearby water bodies.

G9.1.20 Marine Works

- (a) General Requirements
 - (i) Marine structure is defined as any temporary or permanent structure constructed in open water, on the foreshore or in rivers where some part or all of the structure is exposed to the water. Marine work is work on or within a marine structure either during its construction or installation or thereafter.
 - (ii) The Contractor shall ensure that all relevant requirements of marine safety legislation are complied with and shall liaise with the appropriate officers of the Maritime and Port Board of Singapore (MPA) for necessary permissions for work being carried out in waters which come under their jurisdiction.
 - (iii) The Contractor shall liaise with the Building and Construction Board (BCA) with respect to the impact of building materials.
 - (iv) The Contractor shall also ensure that all relevant requirements of environmental legislation are complied with and shall liaise with the appropriate departments of National Environment Agency (NEA).
 - (v) Navigational aids (inclusive of lights and beacons) shall be fitted at appropriate positions as required by marine safety legislation and they shall be maintained in efficient working order. Navigational and other warning lights shall function throughout all hours of darkness or of reduced visibility.
 - (vi) Pre and post condition sonar surveys with runs at 5.0 metre intervals or closer shall be carried out by a hydrographic surveyor approved by MPA. Results of the survey shall be submitted to the S.O. and MPA. Any high spots, levels or sunken object found in the post condition sonar surveys that were not observed in the pre-condition sonar survey shall be removed to the satisfaction of the MPA.
 - (vii) Before starting of any marine works and upon its completion, the Contractor shall ensure appropriate notices to mariners and other port users are issued by the Port Master's Department of the MPA including the payment of necessary fees.
 - (viii) The Contractor shall ensure that lifting machines inclusive of barge mounted cranes, lifting appliances and lifting gears on board of any vessel used for lifting operations in connection with this Contract shall comply with the "lifting operation" requirements of the health and safety specifications.

- (ix) All floating barge / pontoon mounted plant shall be secured against accidental displacement and the barge / pontoon anchored to minimise movement due to the drag of the current etc.
 - (x) At night, the waters surrounding any barge / pontoon in which any form of work is being carried out shall be effectively illuminated to a surface distance of 10 metres away.
 - (xi) All reasonable measures shall be taken by the Contractor to prevent workers from falling into the water. All persons working near the edge of docks, wharves or similar structures, or on board a vessel or barge mounted crane shall be provided with and shall wear a suitable life jacket.
 - (xii) Where any work is being carried out on or near water suitable life buoys with sufficient length of life line shall be placed at strategic points for rescue purposes. The Contractor shall provide a standby emergency boat at a location approved by the S.O..
 - (xiii) The Contractor shall ensure that vertical ladders which afford a means of escape for a person from the water to a safe place are fixed to the outer surface of any marine structure or vessel on which work is being carried out and maintained in an acceptable condition.
- (b) Design, Installation and Dismantling of Marine Structures
- (i) The design of the marine structure shall take into account the maximum possible tidal range and flow, effects of flooding or other works upriver if appropriate, the wash caused by passing vessels and predicted wave height in the worst foreseeable weather conditions. In determining these maximum water levels, the Contractor shall consult relevant tidal charts and tables for the intended location of the marine structure.
 - (ii) The design of marine structures should also take into account the impact load due to the berthing force of vessels, barges, etc.
 - (iii) The Contractor shall ensure that the marine structure is designed by a QP and such design shall be subject to approval by both NEA and MPA. The QP shall supervise its construction and certify the structure as being in compliance with his design and calculations before the marine structure is first taken into use.
 - (iv) The QP shall specify the method by which the marine structure is to be located in position, if it is pre-fabricated, and shall supervise closely its being placed there. This work shall be carried out by persons having adequate previous experience of similar work, or by persons under the close control of supervisors who have such experience. The marine works shall be supervised by a dedicated and qualified supervisor.
 - (v) If assembled in-situ, only competent workers are to be used, and appropriate precautions must be taken to ensure the safety of those persons engaged in such work.
 - (vi) Alterations or modifications proposed to the marine structure shall be planned and designed by a QP subject to the approval of the S.O.. The QP shall subsequently certify that such works have been carried out in accordance with his designs and calculations before being taken back into use.

- (vii) A method statement containing relevant safety requirements shall be generated by a QP together with a risk assessment for the installation, dismantling and subsequent removal from site of the marine structure and this shall be approved by the S.O. before any such work takes place.
 - (viii) All temporary marine structures erected shall be dismantled and removed on completion of the works.
 - (ix) Construction of marine works shall comply with requirements for erosion control and turbidity curtains for protection of the marine environment during construction.
- (c) Working Platforms
- (i) Where a person is required to work above water, proper working platforms in compliance with Scaffold Regulations shall be provided. They shall be maintained in position along all open sides of the platforms and any associated means of access except the parts where it is necessary for this edge protection to be removed for the work in progress.
 - (ii) If required by the S.O. safety nets shall be placed under any working platform over water, at a sufficient height to prevent a person coming into contact with the water taking into account the maximum tidal range.
 - (iii) No-one shall carry out any work over water from a temporary work platform unless he is wearing a safety harness which is secured by a lanyard to a fixed point or lifeline which has first been inspected and authorised by a Safety Supervisor as safe for use. Such authorisation shall be recorded in writing in a register kept for that purpose.
 - (iv) Barges and pontoons and similar which are used as working platforms shall be properly constructed and sufficiently stable to avoid tipping. All persons working on them shall wear appropriate buoyancy aids and be instructed in what action to take in the case of any capsize or man overboard. Each shall have an appropriately stocked first aid box on board.
- (d) Water Transport
- (i) Persons may only be transported across water to their work locations in vessels certified by MPA. All persons carried on board as passengers must be seated throughout the crossing and the vessel may only carry the approved number of passengers. If materials are carried in addition to passengers, the vessel must not be overloaded. Operators of these vessels shall hold a valid MPA license to operate.
 - (ii) The vessel must be equipped with the scheduled number of suitable life-saving appliances according to its capacity and these should be maintained in a satisfactory condition.
 - (iii) No smoking shall be permitted on board the vessel.
 - (iv) The embarkation and disembarkation points for passengers at the land side and at the work location shall be fitted with hand rails and mid rails and shall be designed to make access safe during all tidal and weather conditions. Grab ropes shall be fitted near the waterline at all such locations and lifelines provided and maintained close to the edge.

- (v) When not engaged in the transportation of passengers the vessel shall be on standby for rescue and other emergency purposes.
 - (vi) The Contractor shall make available a vessel together with operator with a seating capacity of not less than six passengers to enable the S.O. site staff to carry out inspections of the marine works at any time during the working hours of the Contract.
 - (vii) All barges, pontoons, vessels and similar used for transporting materials to the work locations for whatever reasons shall be certified by MPA. All persons who work on board shall be subject to the site rules and regulations of the Contractor and shall take part in such safety activities, co-ordination meetings and briefings as are required under the Contractor's Safety Management System.
 - (viii) All sunken vessels, barges, pontoons that may have sunk during the project shall be salvaged and removed.
- (e) Inspection and Remedial Works
- (i) The marine structure shall be inspected by a competent person before any person carries out work in it at the start of any shift or, if continuous working is undertaken, at least once in every period of 12 hours. The results of such inspection shall be recorded in writing and the entry countersigned by the Project Manager or his nominated representative on a daily basis.
 - (ii) The marine structure shall also be inspected by a competent person after any incident which may have affected its strength or structural integrity.
 - (iii) If any remedial works are found to be necessary all workers not involved in such work shall be removed to a safe place on land before such work is commenced.
 - (iv) Any remedial works shall be designed by a PE and approved by the S.O..
- (f) Diving
- (i) All diving works shall comply with "SS 511, Diving at Work". Prior to the commencement of any dive, the Contractor shall submit a dive plan in accordance to SS 511 for the S.O.'s approval.
- (g) Emergency Procedures
- (i) The Contractor shall draw up emergency plans to cover all foreseeable contingencies and these shall be approved by the S.O. before any marine work takes place. They shall include provision for the setting up of an emergency control centre and management team to co-ordinate the rescue procedure or other response.
 - (ii) The Contractor shall ensure that all persons including those who are not employees of the Contractor or of his sub-contractors who work on or over water for whatever reason are briefed in these procedures, in appropriate languages which they understand, before commencing work.
 - (iii) A desk top exercise shall be held for the worst case emergency scenario involving the marine structures within 3 months of the permanent works commencing thereon.

END OF SECTION